

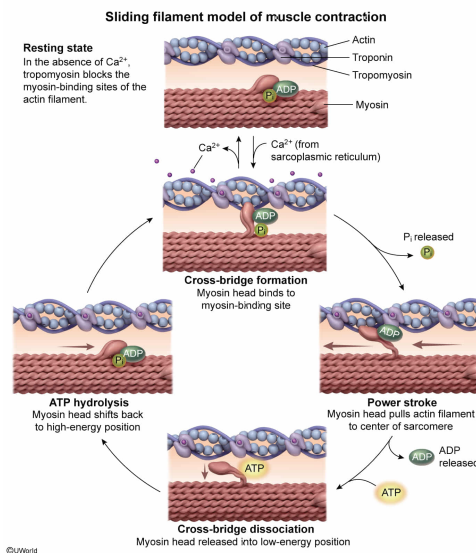
During muscle contraction, the power stroke is the pivoting motion of the myosin head that pulls the actin filament inward to shorten the sarcomere. Given this, which of the following events most immediately precedes this power stroke motion?

- ☐ A. The binding of ATP to the myosin heads [23%]
- ☐ B. The hydrolysis of myosin-bound ATP [28%]
- ✓ ☒ C. The release of P_i from the myosin head [34%]
- ☐ D. The binding of Ca^{2+} to tropomyosin [12%]

Correct

35% answered correctly

Explanation:



Muscle contraction is dependent on the interaction between the myosin (thick) and actin (thin) protein filaments in the **sarcomere**, the basic contractile unit of muscle tissue. To generate muscle contraction, the sarcomere functions as follows:

1. When the muscle fiber is at rest, the myosin head is in its high-energy conformation (upright and bound to ADP and P_i), and the actin filaments are bound by the regulatory proteins tropomyosin and troponin. Tropomyosin is an elongated protein that wraps around the actin filament to block myosin-binding sites on this thin filament. Troponin is a small protein complex associated with tropomyosin.
2. Following a **depolarizing stimulus** by a motor neuron, Ca^{2+} ions are released from the sarcoplasmic reticulum into the cytosol. The abundant cytosolic Ca^{2+} ions *bind troponin*, causing a conformational change that ultimately pulls on tropomyosin and exposes the myosin binding sites on the actin filaments (**Choice D**).
3. When the active sites are exposed, the myosin head is able to bind strongly to the actin filament, forming a cross-bridge.
4. The **dissociation of P_i** initiates the power stroke, which is the actual pivot of the actin-bound myosin head that drags the actin filament toward the center of the sarcomere. This directly results in **shortening of the sarcomere**. The release of ADP from the myosin head occurs at the end of the power stroke. The myosin head is now in its low-energy conformation.
5. A new ATP molecule binds the myosin head and the cross-bridge disassembles (**Choice A**).

6. Hydrolysis of the ATP molecule allows the myosin head to shift back into its upright, high-energy conformation in preparation for a new cycle of contraction (**Choice B**).
7. The cycle of cross-bridge formation and disassembly continues until motor neuron signaling ceases, and Ca^{2+} is sequestered back into the sarcoplasmic reticulum.

Educational objective:

When ATP bound to a myosin head hydrolyzes, the myosin head shifts back to the high-energy position. Binding of calcium to troponin then allows the myosin head to attach to the actin filament. Dissociation of P_i leads to a power stroke, which shortens the sarcomere. After the release of ADP, a new ATP molecule binds to the myosin head, causing it to dissociate from the actin filament. The action of many myosin heads cycling through this process leads to muscle contraction.

Time spent:0 sec

Subject:

Biology

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System:

Musculoskeletal System